

**UNITED UNIVERSITY RAWATPUR, PRAYAGRAJ, UTTAR
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**FACULTY OF COMPUTER APPLICATIONS
SESSION 2025 – 2026**

**A PROJECT REPORT
ON
E-Commerce Sales Analysis and Visualization Using Python**

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Mini Project

E-Commerce Sales Analysis and Visualization **Using Python**

Problem Statement

The rapid growth of the e-commerce industry has led to the generation of large amounts of sales data from different products, customers, and regions. However, raw data alone is not enough for effective business decision-making. Without proper analysis and visualization, it becomes difficult to identify sales trends, customer behavior, and profitable products.

This project uses Python libraries such as Pandas, Matplotlib, Seaborn, and Plotly to analyze and visualize e-commerce sales data. The analysis helps in understanding sales patterns, product performance, customer behavior, and the impact of pricing and quantity on overall sales.

Objectives

1. Load and inspect the dataset.
2. Perform statistical analysis on sales data.
3. Visualize sales distribution using histogram.
4. Analyze category-wise sales performance using boxplot.
5. Study the relationship between Quantity and Total Sale using scatter plot.
6. Generate correlation heatmap for important features.
7. Derive business insights and recommendations.

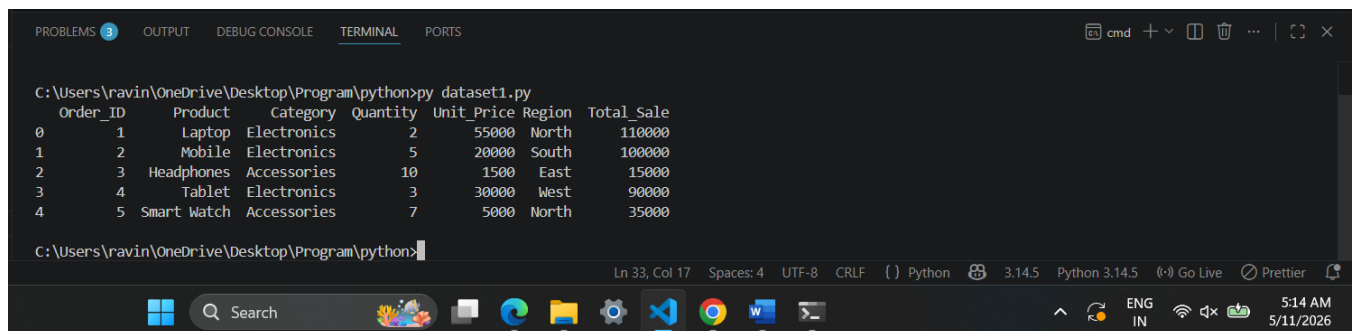
Step 1: Import Necessary Libraries

```
python > step1.py
1 # Import libraries
2 import pandas as pd
3 import seaborn as sns
4 import matplotlib.pyplot as plt
5 import plotly.express as px
6
7 # Style setting
8 sns.set(style='darkgrid')
```

Step 2: Create Dataset

```
dataset1.py > ...
1 import pandas as pd
2 import seaborn as sns
3 import matplotlib.pyplot as plt
4 import plotly.express as px
5
6 # Create E-Commerce Sales Dataset
7
8 data = {
9     'Order_ID': [1,2,3,4,5,6,7,8,9,10],
10
11     'Product': ['Laptop', 'Mobile', 'Headphones', 'Tablet', 'Smart Watch',
12               | 'Keyboard', 'Monitor', 'Printer', 'Mouse', 'Camera'],
13
14     'Category': ['Electronics', 'Electronics', 'Accessories', 'Electronics',
15                | 'Accessories', 'Accessories', 'Electronics', 'Electronics',
16                | 'Accessories', 'Electronics'],
17
18     'Quantity': [2,5,10,3,7,8,2,1,12,2],
19
20     'Unit_Price': [55000, 20000, 1500, 30000, 5000, 1200, 15000, 10000, 800, 45000],
21
22     'Region': ['North', 'South', 'East', 'West', 'North',
23              | 'South', 'East', 'West', 'North', 'South']
24 }
25
26 # Convert into DataFrame
27 df = pd.DataFrame(data)
28
29 # Create Total Sale column
30 df['Total_Sale'] = df['Quantity'] * df['Unit_Price']
31
32 # Display first rows
33 print(df.head())
```

Output –



```
C:\Users\ravin\OneDrive\Desktop\Program\python>py dataset1.py
Order_ID  Product  Category  Quantity  Unit_Price  Region  Total_Sale
0         1       Laptop    Electronics    2      55000   North    110000
1         2       Mobile    Electronics    5      20000   South    100000
2         3   Headphones  Accessories   10       1500   East     15000
3         4       Tablet    Electronics    3      30000   West     90000
4         5   Smart Watch  Accessories    7       5000   North     35000

C:\Users\ravin\OneDrive\Desktop\Program\python>
```

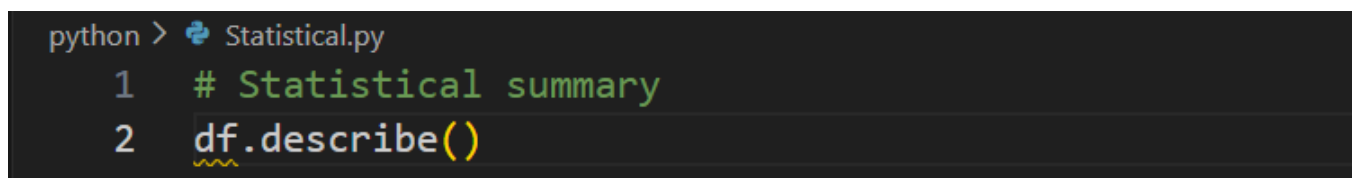
Step 3: Data Exploration

Dataset Information



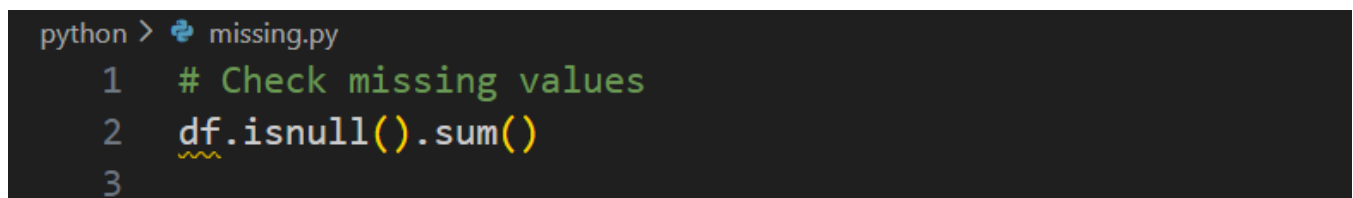
```
python > dataset.py
1  # Dataset information
2  df.info()
3
```

Statistical Summary



```
python > Statistical.py
1  # Statistical summary
2  df.describe()
```

Check Null Values



```
python > missing.py
1  # Check missing values
2  df.isnull().sum()
3
```

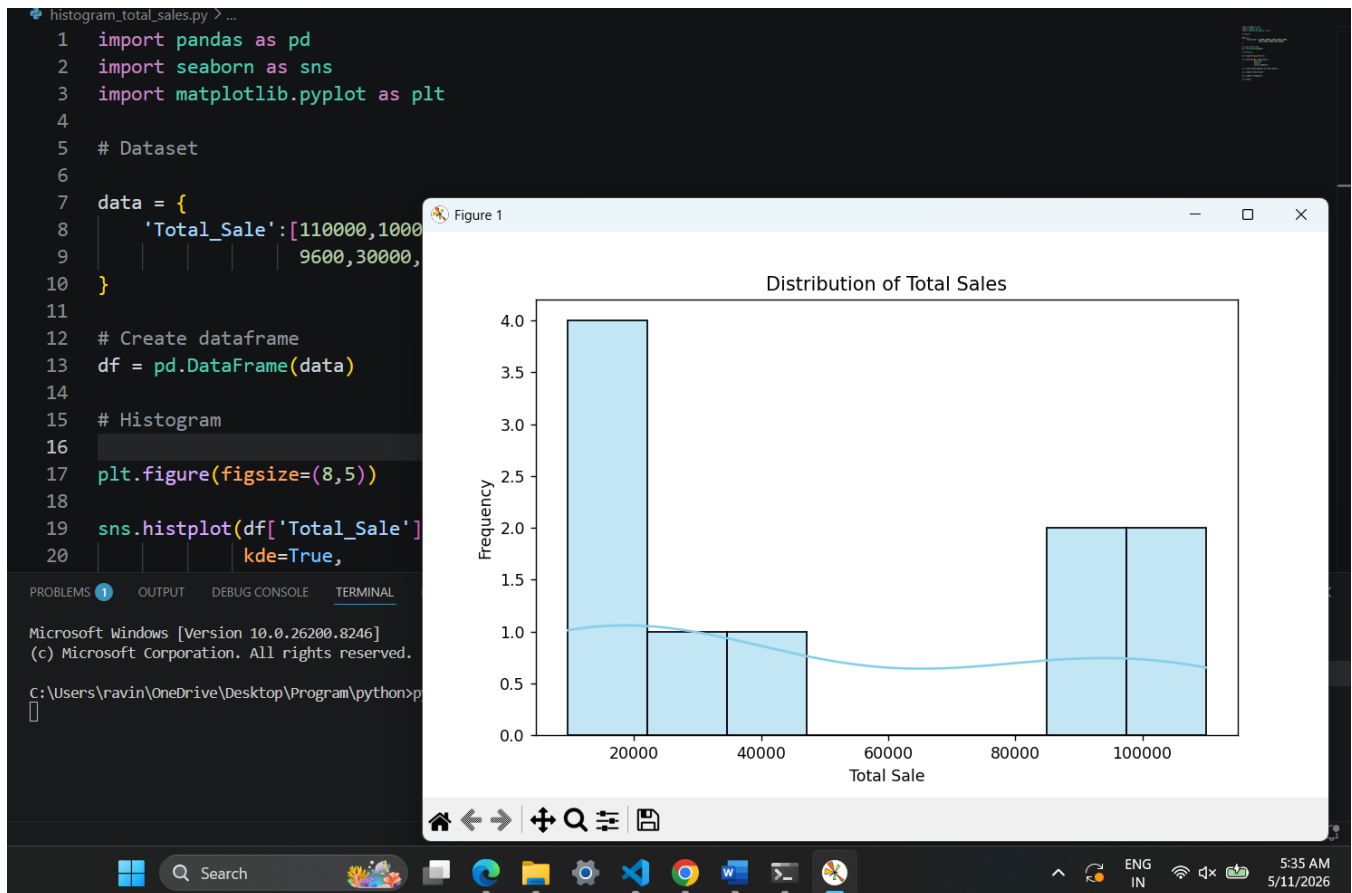
List Columns

```
python > list_col.py
1 # List all columns
2 df.columns.tolist()
```

Step 4: Data Visualization

```
histogram_total_sales.py > ...
1 import pandas as pd
2 import seaborn as sns
3 import matplotlib.pyplot as plt
4
5 # Dataset
6
7 data = {
8     'Total_Sale': [110000, 100000, 15000, 90000, 35000,
9                  9600, 30000, 10000, 9600, 90000]
10 }
11
12 # Create dataframe
13 df = pd.DataFrame(data)
14
15 # Histogram
16
17 plt.figure(figsize=(8,5))
18
19 sns.histplot(df['Total_Sale'],
20             kde=True,
21             bins=8,
22             color='skyblue')
23
24 plt.title('Distribution of Total Sales')
25
26 plt.xlabel('Total Sale')
27
28 plt.ylabel('Frequency')
29
30 plt.show()
31
32
```

Output :-



Observation

- The histogram shows that most sales are concentrated in medium and high-value ranges.
- Few products generate extremely high revenue.

Business Meaning

- Electronics products contribute major revenue.
- Accessories provide smaller but regular sales.

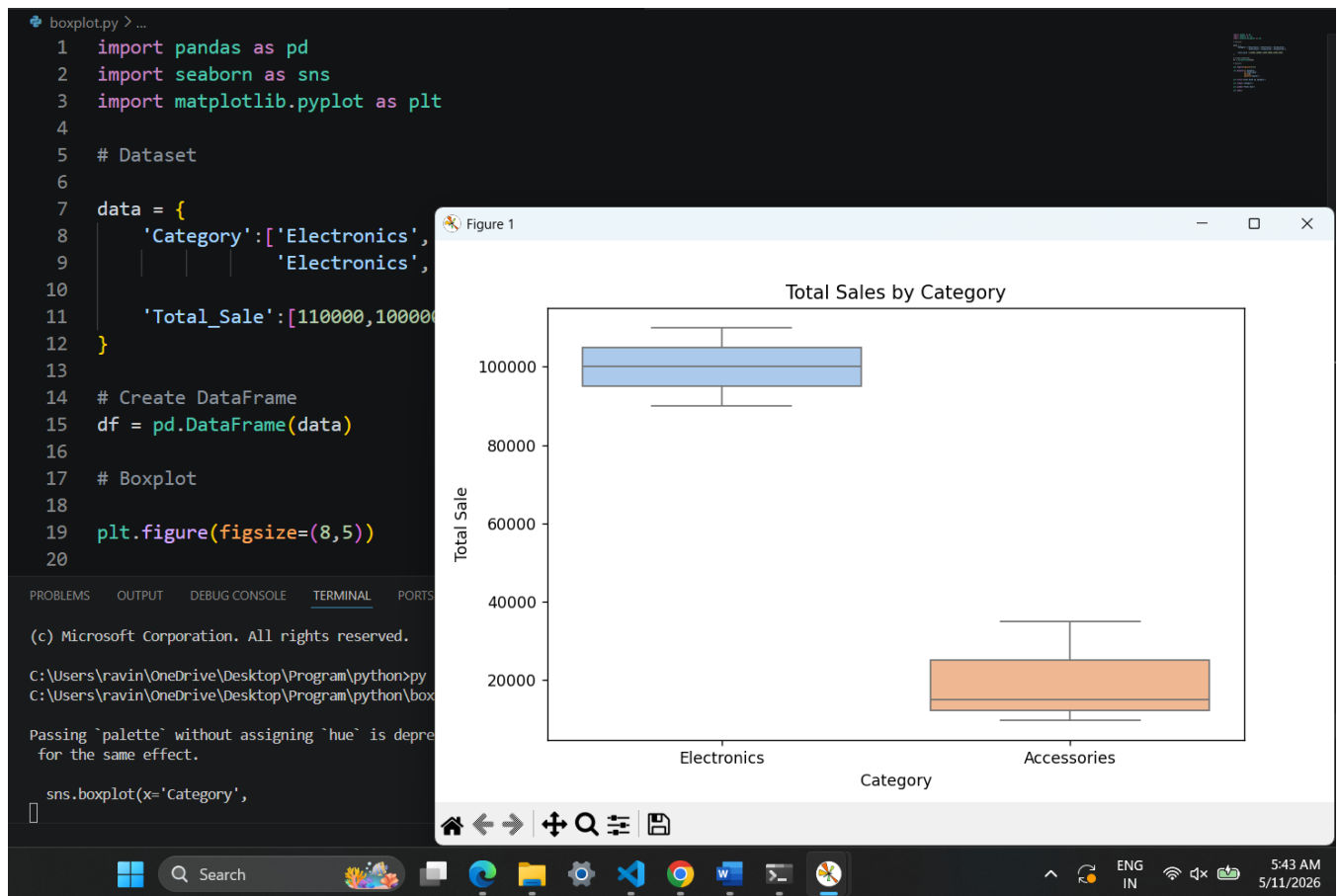
02. Boxplot of Sales by Category

```
boxplot.py > ...
1  import pandas as pd
2  import seaborn as sns
3  import matplotlib.pyplot as plt
4
5  # Dataset
6
7  data = {
8      'Category': ['Electronics', 'Electronics', 'Accessories',
9                  'Electronics', 'Accessories', 'Accessories'],
10     'Total_Sale': [110000, 100000, 15000, 90000, 35000, 9600]
11 }
12
13
14 # Create DataFrame
15 df = pd.DataFrame(data)
16
17 # Boxplot
18
19 plt.figure(figsize=(8,5))
20
21 sns.boxplot(x='Category',
22             y='Total_Sale',
23             data=df,
24             palette='pastel')
25
26 plt.title('Total Sales by Category')
27
28 plt.xlabel('Category')
29
```

```
29
30 plt.ylabel('Total Sale')
31
32 plt.show()
```

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Output :-



Observation

- Electronics category has higher sales values.
- Accessories show lower but more stable revenue.

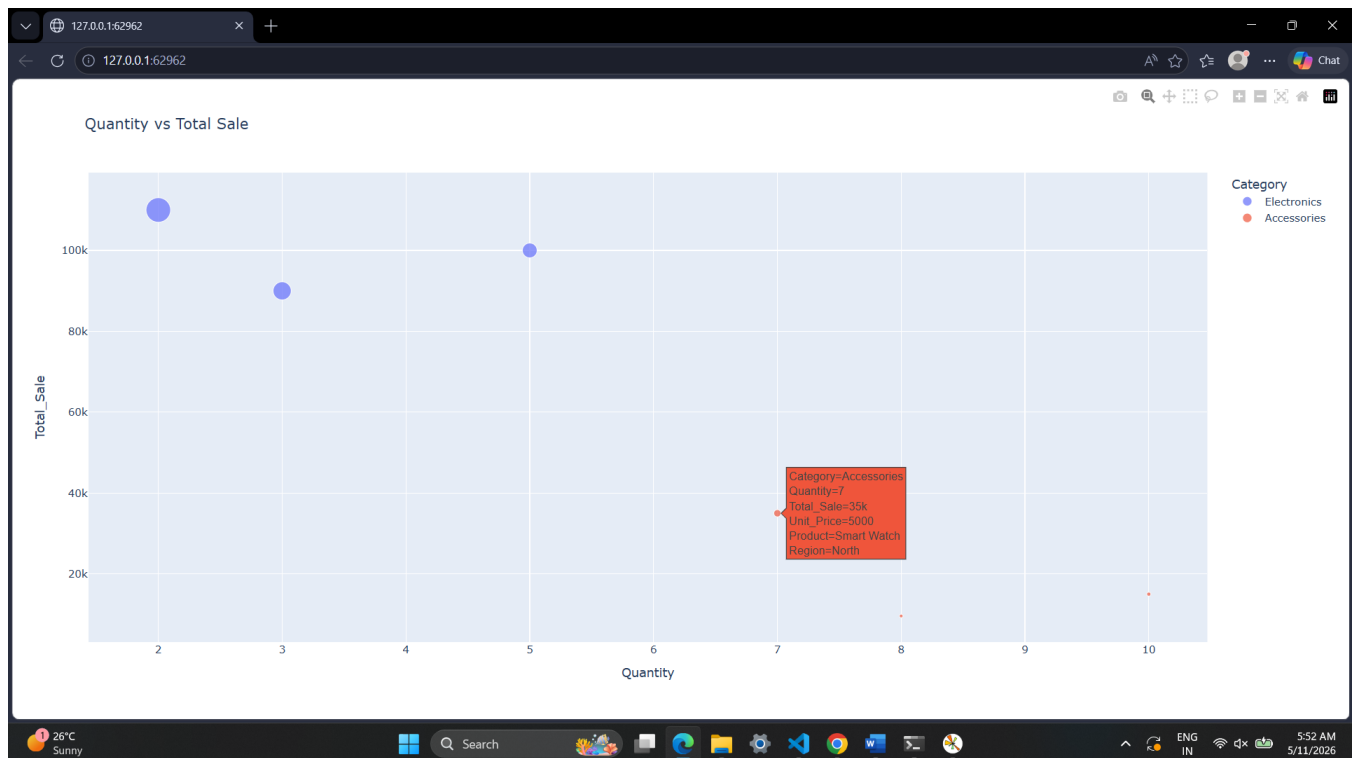
Business Meaning

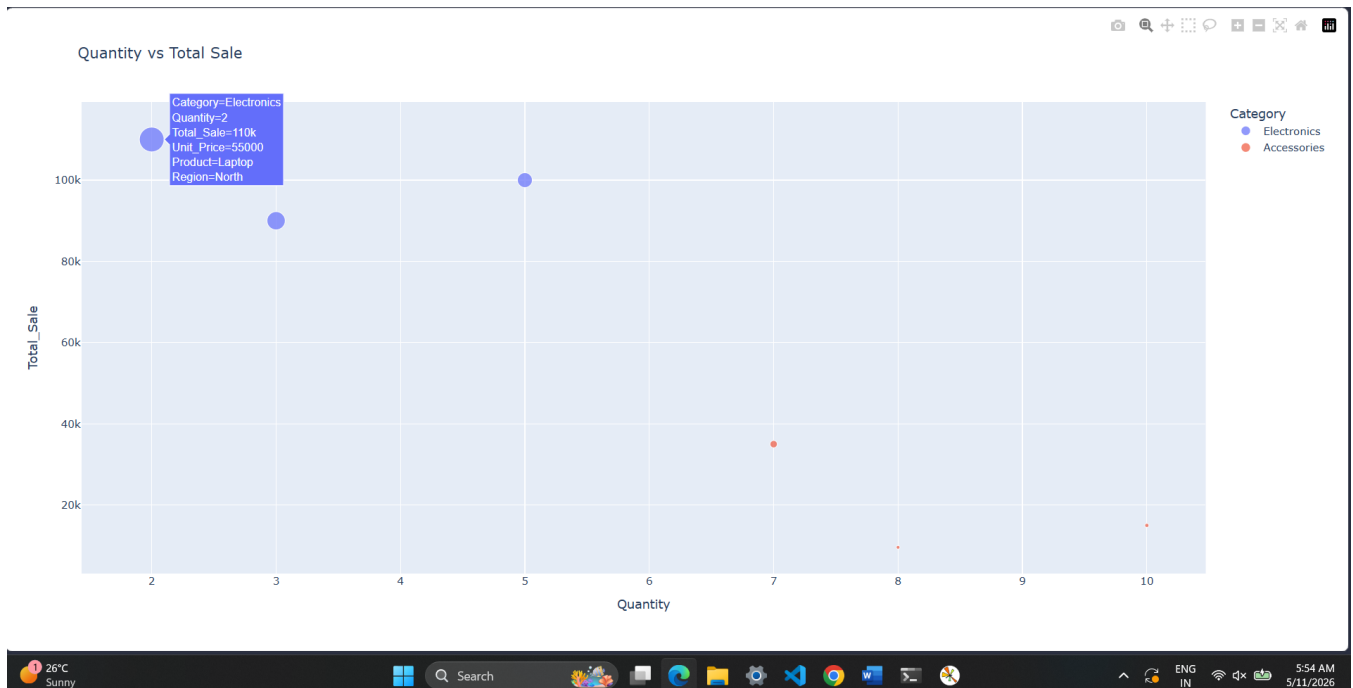
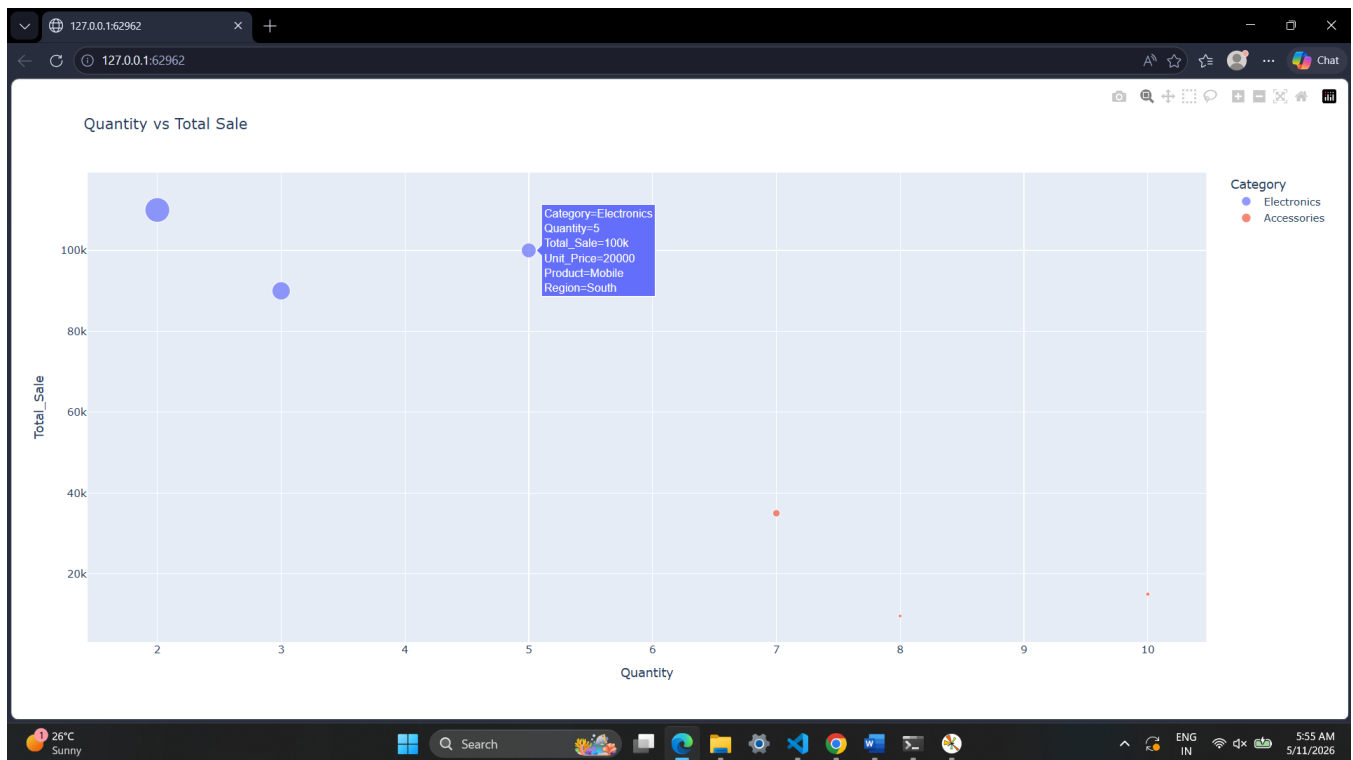
- Electronics are high-profit products.
- Accessories can be used for consistent daily sales.

3. Scatter Plot (Interactive)

```
scatter_plot.py > ...
1  import pandas as pd
2  import plotly.express as px
3
4  # Dataset
5
6  data = {
7
8      'Product': ['Laptop', 'Mobile', 'Headphones',
9                 'Tablet', 'Smart Watch', 'Keyboard'],
10
11      'Category': ['Electronics', 'Electronics', 'Accessories',
12                  'Electronics', 'Accessories', 'Accessories'],
13
14      'Quantity': [2, 5, 10, 3, 7, 8],
15
16      'Unit_Price': [55000, 20000, 1500, 30000, 5000, 1200],
17
18      'Region': ['North', 'South', 'East',
19                'West', 'North', 'South']
20  }
21
22  # Create DataFrame
23  df = pd.DataFrame(data)
24
25  # Create Total Sale column
26  df['Total_Sale'] = df['Quantity'] * df['Unit_Price']
27
28  # Scatter Plot
29
29
30  fig = px.scatter(df,
31                  x='Quantity',
32                  y='Total_Sale',
33                  color='Category',
34                  size='Unit_Price',
35                  hover_data=['Product', 'Region'],
36                  title='Quantity vs Total Sale')
37
38  fig.show()
39
40
41
42
43
44
45
```

Output :-





Observation

- Higher quantity usually increases total sales.
- Some expensive products generate high sales even with low quantity.

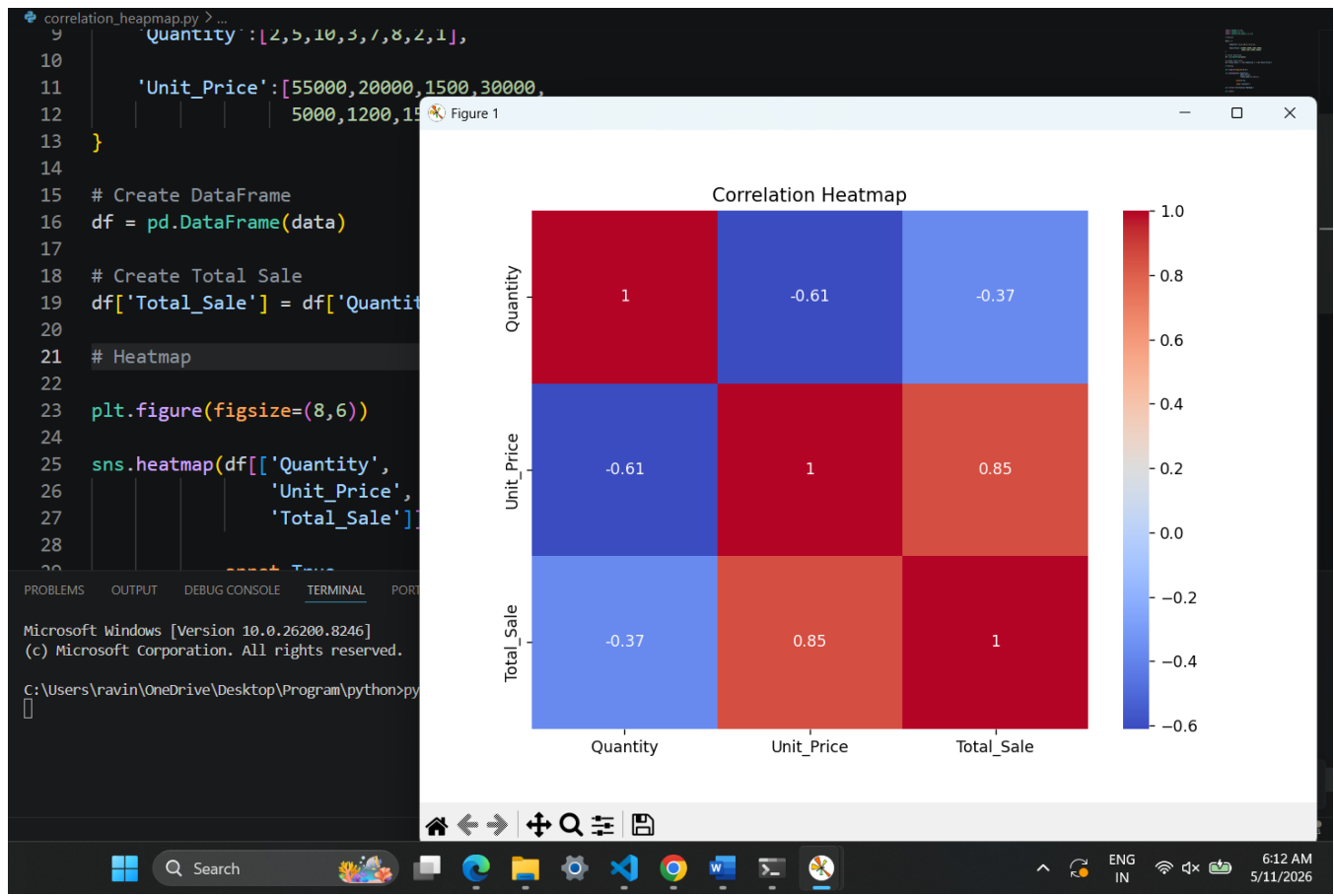
Business Meaning

- Bulk purchasing increases revenue.
- Premium products can generate profit quickl

4. Correlation Heatmap

```
correlation_heatmap.py > ...
1  import pandas as pd
2  import seaborn as sns
3  import matplotlib.pyplot as plt
4
5  # Dataset
6
7  data = {
8
9      'Quantity': [2, 5, 10, 3, 7, 8, 2, 1],
10
11      'Unit_Price': [55000, 20000, 1500, 30000,
12                   5000, 1200, 15000, 10000]
13  }
14
15  # Create DataFrame
16  df = pd.DataFrame(data)
17
18  # Create Total Sale
19  df['Total_Sale'] = df['Quantity'] * df['Unit_Price']
20
21  # Heatmap
22
23  plt.figure(figsize=(8,6))
24
25  sns.heatmap(df[['Quantity',
26                 'Unit_Price',
27                 'Total_Sale']].corr(),
28
29              annot=True,
30
31              cmap='coolwarm')
32
33  plt.title('Correlation Heatmap')
34
35  plt.show()
```

Output :-



Observation

- Strong positive relationship between:
 - Quantity and Total Sale
 - Unit Price and Total Sale

Business Meaning

- Revenue depends on both product price and quantity sold.

Step 5: Key Insights and Business Decisions

1. Sales Distribution

- Most sales come from Electronics products.
- Accessories contribute stable revenue.

2. Product Performance

Electronics

- High revenue
- High profit margin

Accessories

- Lower revenue
- Frequent sales

3. Customer Behavior

- Customers buy accessories frequently.
- Expensive products generate major revenue.

4. Correlation Insight

- Quantity and Unit Price both strongly affect Total Sale.

Recommendations

1. Focus on Electronics

- Increase promotion of:
 - Laptop
 - Mobile
 - Camera

2. Encourage Bulk Orders

Provide:

- Combo offers
- Discount coupons
- Festival sales

3. Improve Customer Retention

- Offer loyalty points
- Provide premium membership

4. Inventory Management

- Maintain stock for fast-selling products.
- Monitor electronics inventory carefully.

Conclusion

This project demonstrates how Python-based data exploration and visualization can convert raw e-commerce sales data into meaningful business insights.

The analysis shows that:

- Electronics products generate maximum revenue.
- Quantity and Unit Price strongly influence Total Sales.
- Data visualization helps businesses make better marketing and inventory decisions.

Using Python libraries like Pandas, Seaborn, Matplotlib, and Plotly makes data analysis easier, interactive, and effective for business growth.